Leveraging Learnable Defense Prompts and One-Step Diffusion Model for Adversarial Purification in Remote Sensing

Weidong Wang, Zhi Li, Long Zheng, Zhenyu Dai, and Chuhua Huang

***Abstract*—Deep neural networks (DNNs) have demonstrated outstanding performance in remote sensing imagery (RSI) analysis. However, their sensitivity to adversarial perturbations remains a critical bottleneck for their deployment in security-sensitive scenarios. Currently, most mainstream adversarial purification methods focus on pixel space example reconstruction, neglecting the effective recovery of deep semantic features and the classification decision space, thereby failing to mitigate the adversarial attacks adequately. To address this challenge, we propose a novel multimodal adversarial purification framework, leveraging learnable defense prompts and a one-step diffusion (LDPs-Diff) model, which aims to achieve robust adversarial defense with semantic consistency. First, we introduce an LDP mechanism in LDP-Diff. Through the learnable optimization of input prompts, LDP-Diff can provide precise semantic guidance for the purification process. Next, we introduce a one-step diffusion model combined with the low-rank adaptation (LoRA) for lightweight fine-tuning, significantly reducing the inference complexity while effectively removing adversarial perturbations and achieving high-quality structural and semantic reconstruction. Moreover, to further enhance the stability of purified examples in the classification space, we propose an entropy-guided logit fusion (EGLF) loss, which quantifies the confidence entropy and adaptively fuses the logit distributions between examples to alleviate the class bias remaining in the classification space after reconstruction. Comprehensive experimental results show that LDP-Diff exhibits excellent defense capabilities in multiple RSI datasets. LDP-Diff achieves an accuracy improvement of 2.40% on the UC Merced (UCM) dataset and realizes a significant performance improvement of 10.84% on the AID dataset compared to the existing state-of-the-art (SOTA) methods. Notably, LDP-Diff exhibits excellent robustness, generalization, computational efficiency, and significant advantages in cross-domain defense.**

***Index Terms*—Adversarial defense, adversarial purification, diffusion model, model robustness, remote sensing image classification.**

I. Introduction

DEEP neural networks (DNNs) [1], owing to their superior data modeling capabilities and highly automated feature extraction, are accelerating the shift toward intelligent spatial and geographic information processing, thereby providing unprecedented technical support for remote sensing imagery (RSI) analysis. In recent years, the application of computer vision techniques in RSI processing has become increasingly pervasive, encompassing key tasks such as scene classification [2], object detection [3], semantic segmentation [4], and super-resolution reconstruction [5]. These advancements have significantly enhanced the ability to perceive and interpret surface information.

As the vulnerability of DNNs to input perturbations becomes increasingly evident, concerns over their deployability and decision reliability have grown significantly. Numerous studies have demonstrated that DNNs are highly vulnerable to adversarial attacks [6]. Attackers can introduce subtle, carefully crafted perturbations into input images, significantly misleading the model's output and causing its predictions to deviate from the ground-truth labels. Although these perturbations are almost imperceptible to the human visual system, they can drastically degrade model performance and even lead to high-confidence mispredictions. In security-sensitive scenarios such as remote sensing scene classification, adversarial attacks can have serious consequences, raising widespread concern in the research community about the security of DNNs [7], [8], [9].

Adversarial attacks are generally classified into white-box attacks and black-box attacks. The former assumes that the attacker has complete access to the model's structure and gradient information, enabling the generation of adversarial examples using gradient-based optimization methods such as fast gradient sign method (FGSM) [10], Carlini–Wagner (CW) [11], and projected gradient descent (PGD) [12]. The latter assumes that the attacker cannot access the model's internal details and instead generates adversarial examples through an input–output query mechanism, typically utilizing transferability assumptions or evolutionary strategies for optimization [13], [14]. A significant body of experimental evidence indicates that even the most advanced classification models exhibit substantial robustness deficiencies when confronted with these attacks [15].

To enhance the defense capability of DNNs against adversarial attacks, researchers have proposed various strategies, primarily including adversarial training [12], adversarial detection [16], [17], and adversarial purification [18], [19], [20], [21]. Adversarial training strengthens robustness by

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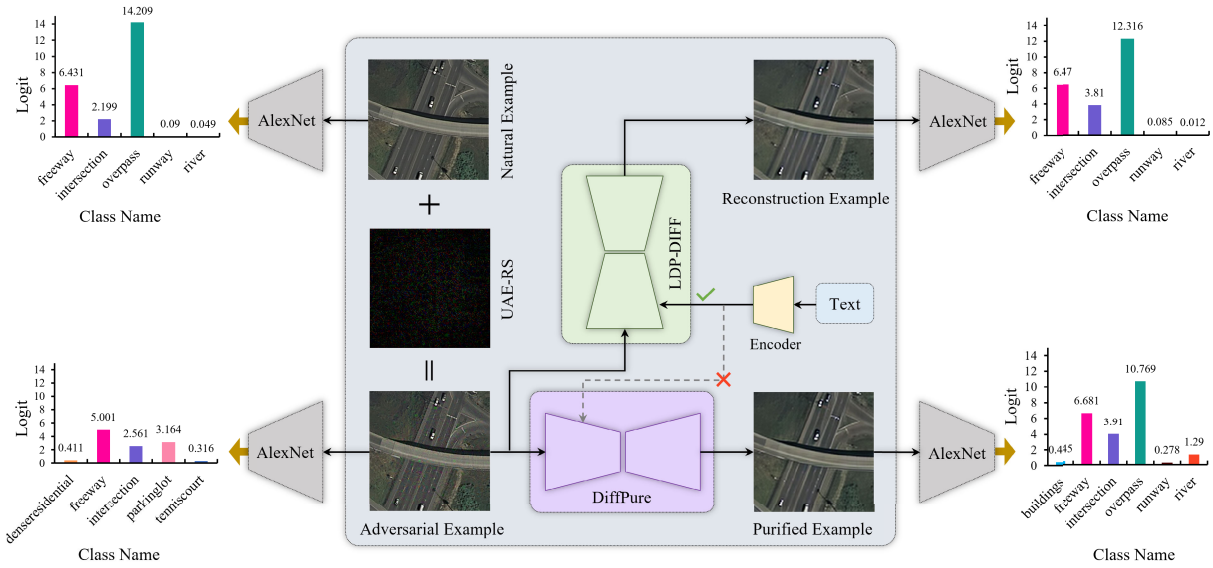


Fig. 1. Visual and semantic space comparison between LDP-Diff and DiffPure on a UCM example. The adversarial example crafted by UAE-RS causes AlexNet to misclassify. DiffPure yields partial recovery with uncertain logits, while LDP-Diff produces high-fidelity reconstruction and restores logits close to those of the natural example.

introducing adversarial examples during training so that the model learns more resilient decision boundaries under perturbations. However, this approach generally incurs substantial computational cost and may suffer from limited generalization to unseen attacks. Adversarial detection aims to identify whether an input is adversarial before the model processes it. This approach is computationally efficient and relatively easy to integrate into existing pipelines. Nevertheless, its robustness is inherently limited, since stronger adaptive attacks can bypass detection mechanisms, thereby reducing reliability in practical scenarios. In contrast, adversarial purification focuses on the inference phase, seeking to restore model performance by removing adversarial disturbances or reconstructing clean representations. Existing purification methods include variational autoencoders (VAEs) [18], [19], generative adversarial networks (GANs) [20], and diffusion models [21]. Yet, many current approaches are constrained to pixel-space reconstruction and fail to sufficiently recover semantic and decision-level consistency. Addressing these shortcomings, our work leverages latent-space diffusion guided by learnable prompts, aiming to achieve more effective adversarial purification for remote sensing tasks.

Diffusion models [22], [23] have gained increasing attention in adversarial purification tasks, mainly due to their strength in modeling complex image distributions. By progressively introducing noise into the data and reconstructing images during the reverse diffusion process, diffusion models can effectively simulate the distribution of natural images and inherently possess the ability to suppress adversarial perturbations. Although the above adversarial purification methods can mitigate the effects of perturbations, they still struggle to eliminate classification bias in reconstruction examples on RSI. As shown in Fig. 1, the toy experiments using the AlexNet [24] classifier on the UC Merced (UCM) [25] dataset reveal that the logit outputs of clean examples exhibit a precise class distribution, allowing the model to perform accurate classification. However, under the universal adversarial examples for remote sensing (UAE-RS) attack [26], the logits corresponding to incorrect classes become markedly amplified, severely distorting the decision boundary and leading to erroneous predictions. Even with methods such as DiffPure, the logit distribution of the reconstruction examples still leans toward the incorrect class, indicating that DiffPure has not fully restored the correct classification perception. This issue is attributed to existing diffusion models being primarily trained on typical datasets, limiting their robustness and adaptability for RSI scenarios.

Recent diffusion-based adversarial purification methods [27], [28] for RSI classification remain confined primarily to pixel-space reconstruction without systematically leveraging the capabilities of multimodal models to enhance semantic representation and improve classification robustness. However, RSI inherently exhibits heterogeneity in object scale, semantic distribution, and background complexity, making unimodal, pixel-level reconstruction insufficient to preserve structural fidelity and semantic consistency. As a result, current adversarial purification approaches for RSI classification continue to face two key challenges. First, they are limited in addressing semantic deviations introduced by adversarial perturbations, as their exclusive focus on pixel-space reconstruction fails to correct decision-space inconsistencies. Second, they fall short in effectively utilizing and integrating multimodal semantic features, thereby restricting potential gains in classification robustness.

Addressing these limitations gives rise to a fundamental research question: How can multimodal information be effectively integrated to construct a robust and semantically aligned adversarial purification framework for RSI classification, and

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thereby overcome the shortcomings of existing unimodal approaches?

Therefore, this article proposes a multimodal adversarial purification framework, learnable defense prompts and a one-step diffusion (LDPs-Diff), which integrates the LDPs mechanism with a one-step diffusion model to enhance adversarial robustness and cross-domain generalization for RSI classification. Specifically, the LDP mechanism learns input prompts that semantically guide the diffusion process, steering example reconstruction toward the target class features. This mechanism significantly strengthens semantic constraints during purification and effectively mitigates semantic bias induced by adversarial perturbations, thereby improving robustness in the decision space. Meanwhile, LDP-Diff employs a computationally lightweight one-step diffusion model to balance semantic reconstruction quality with inference efficiency and incorporates the low-rank adaptation (LoRA) [29] to enable low-cost, parameter-efficient modulation of the diffusion process. Furthermore, an entropy-guided logit fusion (EGLF) loss is proposed to fuse the semantic prompt features from LDP adaptively. By leveraging entropy-based uncertainty estimation, EGLF dynamically adjusts the strength of semantic guidance within the decision space, effectively correcting residual classwise bias and improving prediction consistency and robustness.

LDP-Diff establishes a synergistic purification framework across pixel and semantic space. As shown in Fig. 1, the reconstruction example exhibits satisfactory visual quality. It produces logit distributions closely aligned with the corresponding natural example, demonstrating that LDP-Diff ensures accurate classification while effectively rectifying logit bias. The main contributions of this work are summarized as follows.

1) To our knowledge, this is the first work to introduce a multimodal adversarial purification framework for RSI classification, termed LDP-Diff. By leveraging LDP with a one-step diffusion model, LDP-Diff achieves an effective balance between fast inference and robust defense against diverse adversarial attacks.
2) We further propose an EGLF loss to enhance semantic consistency and decision-space robustness during the diffusion process. By leveraging entropy-based uncertainty estimation, EGLF adaptively fuses semantic prompt features and effectively mitigates the risk of misclassification.
3) Extensive experiments conducted across multiple RSI datasets demonstrate that LDP-Diff consistently outperforms state-of-the-art (SOTA) methods regarding classification accuracy, generalization ability, and cross-domain robustness. These results confirm the proposed framework's broad applicability and high practical value in security-sensitive remote sensing scenarios.

II. Related Works

A. Problem Formulation

Before introducing related works, we briefly describe the standard formulation of RSI classification. Given an RSI dataset $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^{N}$, where each input example $x_i \in \mathbb{R}^{H\times W\times C}$ has height H, width W, and C channels, and the corresponding label $y_i \in \{0, 1, \ldots, K-1\}$ denotes one of K possible classes, the goal of RSI classification is to learn a model $f_\theta : \mathbb{R}^{H\times W\times C} \to \mathbb{R}^K$, where $f_\theta(x) \in \mathbb{R}^K$ denotes the logit vector, i.e., the unnormalized prediction scores for each class produced by the final fully connected layer. The model is typically optimized using the cross-entropy loss, defined as follows:

$$\mathcal{L}(\theta) = \mathbb{E}_{(x,y)\sim\mathcal{D}}\left[-\log\frac{\exp\left(f_\theta(x)_y\right)}{\sum_{i=1}^{K}\exp\left(f_\theta(x)_i\right)}\right] \tag{1}$$

where $f_\theta(x)_y$ denotes the logit corresponding to the ground-truth class y. $\mathcal{L}(\theta)$ quantifies the discrepancy between the predicted class distribution and the true distribution; minimizing it effectively enhances the classification accuracy of the model.

In contrast, adversarial attacks aim to generate adversarial example x' that causes the model f_θ to misclassify. Typically, for a given clean example x, the adversarial example x' can be represented as follows:

$$x' = x + \delta, \quad \text{s.t.} \quad \|\delta\| \le \epsilon \tag{2}$$

where δ denotes the adversarial perturbation, and ϵ specifies a norm-bound constraint to ensure that the adversarial example x' remains visually similar to the original input x.

B. Adversarial Attack

1) Classic Adversarial Attack: FGSM [10] is a classical adversarial attack that generates adversarial example x' by perturbing the input example x along the direction of the gradient's sign. It is formulated as follows:

$$x' = x + \epsilon\cdot\operatorname{sign}\left(\nabla_x\,\mathcal{L}(\theta, x, y)\right) \tag{3}$$

where $\operatorname{sign}(\cdot)$ is the sign function, and $\nabla_x\,\mathcal{L}(\theta, x, y)$ denotes the gradient of the loss with respect to the input x. The iterative FGSM (IFGSM) [30] extends FGSM by applying the perturbation iteratively with a smaller step size α

$$x'_{t+1} = \operatorname{Clip}_{x,\epsilon}\left(x'_t + \alpha\cdot\operatorname{sign}\left(\nabla_{x'_t}\,\mathcal{L}\left(\theta, x'_t, y\right)\right)\right) \tag{4}$$

where x'_t is the adversarial example at iteration t, and $\operatorname{Clip}_{x,\epsilon}(\cdot)$ ensures that x'_{t+1} remains within an ϵ-ball around the input x'_t.

The CW attack [11] is proposed to effectively overcome the defense distillation. It generates the adversarial example x' by solving the following optimization problem:

$$\min_{\delta}\|\delta\|_p + c\cdot f_\theta(x+\delta, y). \tag{5}$$

Here, c is the hyperparameter that controls the balance of the perturbation. CW attack produces highly imperceptible perturbations but requires high computational costs.

The PGD attack [12] is an iterative attack similar to IFGSM but projects each update back to the allowed perturbation set

$$x'_{t+1} = \prod_{\mathcal{B}(x,\epsilon)}\left(x'_t + \alpha\cdot\operatorname{sign}\left(\nabla_{x'_t}\mathcal{L}\left(\theta, x'_t, y\right)\right)\right) \tag{6}$$

where $\prod_{\mathcal{B}(x,\epsilon)}(\cdot)$ denotes projection onto the ϵ-ball $\mathcal{B}$ centered at x.

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The trade-off PGD (TPGD) attack [31] generates adversarial perturbations by calculating the Kullback–Leibler (KL) divergence between the probability distributions of the predictions for adversarial and clean examples

$$x'_{t+1} = x'_t + \alpha \cdot \mathrm{KL}\left(\nabla_{x'_t} \mathcal{L}\left(\theta, x'_t, y\right)\right). \tag{7}$$

The Jitter attack [32] is a geometry-based adversarial attack that generates adversarial example by applying random geometric transformations to the input. Unlike gradient-based attacks, it manipulates spatial features to influence classification decisions without relying on gradient information.

2) Adversarial Attack for RSI: Researchers have proposed a range of more tailored adversarial attack methods in response to the adversarial attacks for RSI. Zhu et al. [33] introduced the feature approximation strategy, which generates adversarial examples by mapping the features of clean images to virtual features that do not belong to any specific category. DeMPAA [34] combined the feature extraction and patch generation module, optimizing the position of numerous small perturbation patches through probability-guided random sampling, achieving high success rates and strong stealthiness. To enhance generalization, Xu and Ghamisi [26] proposed UAE-RS attack. This method includes two variants: Mixup, which perturbs features to improve cross-model transferability, and Mixcut, which enhances adversarial strength through image slice mixing. In object detection, Lian et al. [35] propose an adaptive-patch-based physical attack (AP-PA) that explicitly accounts for dynamics and scale variations, allowing patches to be placed adjacent to rather than occluding the target. The contextual background attack (CBA) [36] advances a background-centric, context-aware attack paradigm in the physical world, providing a systematic reference for background-based adversarial strategies.

C. Adversarial Purification

1) General Adversarial Purification: Applying GANs and VAEs in adversarial purification has gained significant attention in recent years. For instance, Defense-GAN [20] and APE-GAN [37] utilize GANs to remove adversarial perturbations by training the network to mitigate their effects. Defense-VAE [18] and MAD-VAE [19] leverage VAEs to model data distributions, enhancing adversarial purification capabilities. Zhou et al. [38] introduced the adversarial noise removal network (ARN), which effectively defends against unseen adversarial examples by utilizing attack-invariant features to recover natural data. Naseer et al. [39] proposed the neural representation purifier (NRP), which eliminates adversarial perturbations independently of the label space and does not require large amounts of training data.

Diffusion models have provided a new approach for adversarial purification tasks. DiffPure [21] utilizes the reverse generative process of diffusion models to solve the stochastic differential equation (SDE) and restore clean images. The guided diffusion model for adversarial purification (GDMP) [40] embeds the purification process into the denoising–diffusion process of denoising diffusion probabilistic models (DDPMs) [22]. Additionally, GNS [41] explores the importance of hyperparameter selection in adversarial purification and proposes a progressive noise scheduling strategy to optimize the purification process.

2) Adversarial Purification for RSI: The aforementioned UAD-RS [27] is a universal adversarial purification framework that mitigates adversarial perturbations via a pretrained diffusion model. Liu et al. [28] proposed an adversarial restoration network, D3R-Net, which converts adversarial purification into Gaussian noise and incorporates a filtering module to preserve semantic information, thereby restoring remote sensing images consistent with the original distribution. Cheng et al. [42] introduced the perturbation search GAN (PSGAN), which models the perturbation distribution in adversarial examples rather than directly modeling the clean example distribution. PSGAN adjusts the seeking radius to balance the diversity of reconstructed examples with optimization stability. Task-guided denoising network (TGDN) [43] trains a denoising network to transform adversarial examples into forms closer to their clean counterparts. DiffDual-AD [44] presents a diffusion-based dual-stage adversarial defense framework for RSIs.

A significant advantage of adversarial purification is that it is a universal module applied to pretrained models. Thus, it avoids the "robustness–accuracy" trade-off, time consumption seen in adversarial training, and poor generalizability in adversarial detection.

D. Diffusion Model

In recent years, diffusion models [22], [45], [46], [47] have made significant breakthroughs in various fields, particularly excelling in image generation, restoration, and adversarial purification tasks. The most representative diffusion model is DDPM [22], which successfully reconstructs high-quality images by progressively predicting noise and performing reverse diffusion. However, DDPM requires multiple iterative calculations during the inference phase, resulting in high computational overhead. To reduce this computational burden, stable diffusion [47] utilizes VAEs to map images to latent space and perform diffusion within this space, significantly lowering computational costs while maintaining high-quality generation capabilities. Furthermore, stable diffusion introduces a text-guided mechanism, embedding linguistic priors into the diffusion network and enhancing semantic control during image generation, which is typically achieved by integrating text encoders such as CLIP or Transformers. One-step diffusion models [45], [46] streamline the traditional multistep denoising process into a single step by utilizing techniques like knowledge distillation and distribution matching, significantly improving inference speed and making them especially suitable for resource-constrained or low-latency applications.

Additionally, LoRA [29], [48] has emerged as a key strategy for enhancing the adaptability of diffusion models. LoRA fine-tunes pretrained models by inserting trainable low-rank matrices, avoiding the computational and storage overhead associated with full model updates. LoRA has been successfully integrated into variants of stable diffusion, demonstrating substantial adversarial purification potential in remote sensing security analysis.

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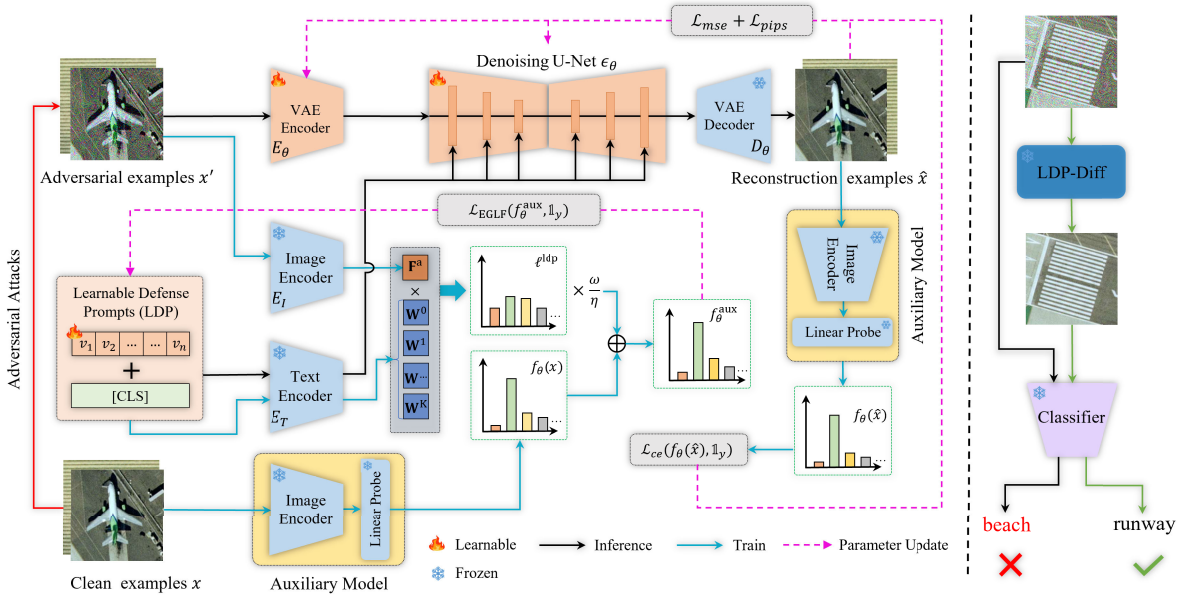


Fig. 2. Overview of the proposed LDP-Diff framework. It consists of LDP, CLIP-based image and text encoders, a VAE-based diffusion module, and a frozen auxiliary model for logit supervision. Only the VAE encoder and denoising network are updated during training, while other components remain frozen. LDPs adaptively guide the diffusion process to purify adversarial examples in a single step, significantly improving efficiency. To further enhance classification robustness, an EGLF loss is introduced to refine predictions. As shown on the right, an adversarial example originally misclassified as "*beach*" is correctly classified as "*runway*" after reconstruction by LDP-Diff.

E. Multimodal Pretraining Model

Multimodal pretraining models integrating visual and textual modalities have shown remarkable potential in enabling generalizable representations across diverse downstream tasks. Contrastive language–image pretraining (CLIP) is a foundational framework for cross-modal understanding and generation by jointly aligning image and language representations in a shared embedding space. To improve adaptability in downstream tasks, CoOp [49] and KgCoOp [50] replace manually crafted textual prompts with learnable prompt vectors, facilitating task-specific tuning without retraining the backbone. These approaches significantly improve the efficiency and flexibility of multimodal models in various application scenarios. As research into CLIP's robustness advances, adversarial prompt tuning (APT) [51] improves its resistance to adversarial interference by learning robust textual prompts. AMU-Tuning [52] enhances CLIP's adaptability and stability in few-shot classification tasks by incorporating auxiliary features and uncertainty fusion. Additionally, the adversarial fine-tuning (AFT) [53] method adjusts the input gradient norm, increasing the sparsity and interpretability of gradient saliency maps.

In summary, these advancements in multimodal pretraining offer critical insights into improving the safety, adaptability, and robustness of vision–language models and provide a theoretical foundation in multimodal adversarial purification for RSI.

III. Methodology

A. Overall Framework

The architecture of LDP-Diff, as shown in Fig. 2, consists of the LDP-based prompt vector V, the CLIP-based image encoder E_I and text encoder E_T, the VAE-based encoder E_θ, the one-step diffusion network ϵ_θ, the decoder D_θ, and an auxiliary model f_θ for computing logits. Notably, during the training process of the LDP-Diff model, only the parameters of V, E_θ, and ϵ_θ are updated, while the parameters of the other modules are frozen. The network architecture details and optimization process of LDP-Diff are described in detail below.

B. Learnable Defense Prompt

The adversarial purification methods based on diffusion models, as discussed, typically rely on fixed input transformations or restoration strategies. However, they often fail to sufficiently account for the variability and complexity of adversarial examples. Inspired by [49], [51], [52], and [53], we propose a dynamic and optimizable prompt framework aimed at enhancing the model's adaptability when confronted with various types of adversarial perturbations. Let the learnable prompt be represented as $V = [v_1, v_2, \ldots, v_n]$, where n is the hyperparameter controlling the length of V. The learnable prompts for k classes can then be represented as $S = [v_1, v_2, \ldots, v_K, s_k]$, where s_k is the embedding representation of the k class names. The text encoder E_T is then used to generate the text features

$$W = (E_T(S)) = [w_1, w_2, \ldots, w_k]^{\mathrm{T}} \in \mathbb{R}^{k \times d} \tag{8}$$

where d is the feature dimension. The generated text embeddings W serve two key functions: First, as additional information to the input image, they adjust the image features and provide more accurate semantic guidance, thereby helping the model better reconstruct natural examples. Second, the text embeddings W play an important role in the calculation of

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EGLF, guiding the model's decisions during the adversarial example purification process. This significantly reduces the impact of adversarial perturbations on the classification result, further enhancing classification performance.

C. One-Step Diffusion Model

Diffusion models generate data by gradually corrupting samples with Gaussian noise and then learning the reverse process to recover clean data. To reduce computational cost, stable diffusion performs this process in the latent space and leverages semantic guidance from the CLIP text encoder E_T, providing conditioning features W for noise prediction. At step t, the denoising network ϵ_θ predicts

$$\hat{\epsilon} = \epsilon_\theta(z_t, t, W). \tag{9}$$

For adversarial purification, we adopt a one-step approximation: instead of iterative denoising, only the final step is applied

$$\hat{z}_0 = \frac{1}{\alpha_T}(z_T - \beta_T \hat{\epsilon}) \tag{10}$$

which significantly improves inference efficiency. Furthermore, to adapt stable diffusion to remote sensing purification while preserving the pretrained backbone, we integrate LoRA modules as lightweight, trainable adapters.

The overall purification pipeline is summarized as follows:

$$\hat{x} = D_\theta\left(\epsilon_\theta\left(E_\theta\left(x'\right), E_T(S)\right)\right) \tag{11}$$

where x' is the adversarial input, S denotes semantic prompts, and D_θ reconstructs the purified image.

D. Auxiliary Model

The auxiliary model is composed of the pretrained CLIP image encoder E_I and a linear probe to provide logit information. E_I is used to extract features from the input image, denoted as $F = E_I(x)$, which are then passed through a logit scale layer for scaling transformation. Subsequently, the features undergo a linear mapping through the linear probe

$$f_\theta(x) = \sigma F + b \tag{12}$$

where σ is the weight of the linear probe, b is the bias term, and $f_\theta(x)$ represents the logit of x. During the LDP-Diff training process, the pretrained f_θ avoids interference with the parameter updates of the diffusion model.

E. Entropy-Guided Logit Fusion

Although diffusion models exhibit powerful image reconstruction capabilities and can effectively eliminate adversarial noise in the pixel space, the reconstruction images may retain residual perturbations, which can interfere with the classifier's decision boundaries, leading to biased or incorrect predictions. To address this issue, the proposed EGLF adaptively calibrates the model's logit outputs in the semantic classification space, effectively mitigating the impact of such residual perturbations.

Specifically, let $F^a = E_I(x') \in \mathbb{R}^{1\times d}$ denote the feature representation of adversarial examples x', obtained through the image encoder E_I. This feature is aligned with the text feature matrix $W \in \mathbb{R}^{k\times d}$ to construct semantic logit via dot-product similarity

$$\ell^{\text{ldp}} = \tau F^a W^{\text{T}} \in \mathbb{R}^{1\times k} \tag{13}$$

where τ is a learnable temperature scaling factor controlling the smoothness of the logit distribution.

Considering that the fusion of logit is susceptible to the model and dataset [52], [53], we introduce the entropy as an uncertainty-aware weighting metric to guide the ℓ^{ldp} fusion process. In information theory, entropy is a well-established measure for quantifying uncertainty in probability distributions. To estimate the uncertainty of the natural example x based on the auxiliary classifier $f_\theta(\cdot)$, we first compute the normalized logit probability distribution via the softmax function, let $f_\theta(x)$ be the logit output of the classifier for input x. The corresponding softmax-normalized probability distribution is given by

$$p(x) = \frac{\exp\left(f_{\theta,k}(x)\right)}{\sum_{i=1}^{K}\exp\left(f_{\theta,i}(x)\right)}. \tag{14}$$

The entropy corresponding to $p(x)$ can be calculated as follows:

$$\mathcal{U}(x) = -\frac{1}{\log K}\sum_{k=1}^{K} p_k(x)\log p_k(x). \tag{15}$$

$\mathcal{U}(x) \in [0,1]$ provides an accurate measure of the auxiliary model's predictive uncertainty on natural examples. Leveraging this uncertainty, an exponential weighting factor is designed to dynamically regulate the fusion intensity of semantic and diffusion logits

$$\omega = \exp(\gamma \cdot \mathcal{U}(x)). \tag{16}$$

Here, γ is a hyperparameter that controls the sensitivity of the fusion weight to the uncertainty level. A higher prediction uncertainty leads to a larger corresponding ω, thereby emphasizing the semantic auxiliary logit's role in guiding the classification decision. The final uncertainty-aware fused output is defined as follows:

$$f_\theta^{\text{aux}} = f_\theta^{\text{clean}} + \frac{\omega}{\eta}\cdot \ell^{\text{ldp}} \tag{17}$$

where η is a balancing factor that regulates the overall fusion strength. Finally, EGLF optimizes the corrected f_θ^{aux} through cross-entropy loss

$$\mathcal{L}_{\text{EGLF}} = -\sum_{i=1}^{K} y_k \log f_{\theta,i}^{\text{aux}}. \tag{18}$$

$\mathcal{L}_{\text{EGLF}}$ is convex and differentiable from an optimization perspective, providing stable gradient signals and facilitating convergence to a robust solution space during training. EGLF compensates for the semantic-level limitations of diffusion-based purification and introduces an information-theoretic mechanism to quantify and adaptively respond to prediction uncertainty. It improves the classifier's resilience against adversarial perturbations and enhances the overall robustness of LDP-Diff.

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F. Overall Training Loss

LDP-Diff needs not only to achieve high-quality reconstruction in the pixel space but also to ensure the stability of classification decisions in the semantic space. Therefore, the training loss consists of two core components: reconstruction-level loss and classification-level loss.

The reconstruction-level loss is composed of mean squared error (mse) loss $\mathcal{L}_{\text{mse}}$ and learned perceptual image patch similarity loss $\mathcal{L}_{\text{PIPS}}$ [54]. $\mathcal{L}_{\text{mse}}$ computes the Euclidean distance between the reconstruction image $\hat{x}$ and the clean image x

$$\mathcal{L}_{\text{mse}} = \frac{1}{N}\sum_{i=1}^{N} \|\hat{x}_i - x_i\|_2^2. \tag{19}$$

$\mathcal{L}_{\text{PIPS}}$ measures perceptual differences in feature space

$$\mathcal{L}_{\text{PIPS}} = \mathbb{E}_x \left[\|\phi(\hat{x}) - \phi(x)\|_2^2 \right]. \tag{20}$$

Here, $\phi(\cdot)$ represents the feature extraction operation in the pretrained VGG [55] network.

The goal of the classification-level loss is to ensure the classification stability of $\hat{x}$ from the perspective of category space. This loss not only includes $\mathcal{L}_{\text{EGLF}}$, which is used to guide the model to better retain category information during adversarial purification, but also combines the cross-entropy loss to further optimize the model's classification accuracy for purified examples, thereby effectively reducing the impact of adversarial perturbations on classification results

$$\mathcal{L}_{\text{CE}} = \mathbb{E}_{(x,y)\sim\mathcal{D}} \left[-\log \frac{\exp\left(f_\theta(\hat{x})_y\right)}{\sum_{j=1}^{K} \exp\left(f_\theta(\hat{x})_j\right)} \right]. \tag{21}$$

In summary, the final overall training loss is as follows:

$$\mathcal{L}_{\text{ALL}} = \mathcal{L}_{\text{mse}} + \xi_1 \mathcal{L}_{\text{PIPS}} + \xi_2 \mathcal{L}_{\text{EGLF}} + \xi_3 \mathcal{L}_{\text{CE}} \tag{22}$$

where ξ_1, ξ_2, and ξ_3 are weighting factors. The LDP-Diff algorithm, as outlined in Algorithm 1, encompasses the complete workflow from adversarial input to reconstructed output.

Algorithm 1 LDP-Diff Purification Framework

Input: x': adversarial RSI; y: ground-truth label; V: learnable defense prompt; E_θ: VAE encoder; ϵ_θ: one-step diffusion network; D_θ: decoder; E_I, E_T: CLIP encoders; f_θ: auxiliary model
Output: purified image $\hat{x}$
Randomly initialize θ $(V, E_\theta, \epsilon_\theta)$; Set iteration counter $t \leftarrow 0$;
while $t <$ *number of training iterations* **do**
  Construct semantic prompts $S = [V, s_y]$ and extract $W = E_T(S)$; Encode $z_0 = E_\theta(x')$ and predict noise $\hat{\epsilon} = \epsilon_\theta(z_T, T, W)$; Reconstruct $\hat{x} = D_\theta(\frac{1}{\alpha_T}(z_T - \beta_T \hat{\epsilon}))$;
  Compute $F^a = E_I(\hat{x})$ and logits $\ell^{ldp} = \tau F^a W^T$;
  Extract $p(x) = \text{softmax}(f_\theta(x))$, entropy $\mathcal{U}(x) = -\frac{1}{\log K}\sum p_k \log p_k$; Fuse $f_\theta^{aux} = f_\theta^{clean} + \frac{\exp(\gamma \cdot \mathcal{U}(x))}{\eta} \cdot \ell^{ldp}$;
  Compute losses $\mathcal{L}_{ALL} = \mathcal{L}_{MSE} + \xi_1 \mathcal{L}_{PIPS} + \xi_2 \mathcal{L}_{EGLF} + \xi_3 \mathcal{L}_{CE}$;
  Update θ $(V, E_\theta, \epsilon_\theta)$ via $\theta \leftarrow \theta - \nabla_\theta \mathcal{L}_{ALL}$;
end
return reconstruction RSI $\hat{x}$;

IV. Experiments

A. Dataset

To evaluate the effectiveness of the proposed LDP-Diff algorithm, we conducted extensive experiments on several representative remote sensing scene classification datasets, including UCM, Aerial Image Dataset (AID) [56], NWPU-RESISC45 (NWPU) [57], and universal adversarial examples in remote sensing (UAE-RS) [26].

1) UC Merced: The UCM dataset, provided by the University of California, Merced, serves as a benchmark for high-resolution remote sensing scene classification. It consists of 21 scene classes, each containing 100 RGB images with a spatial resolution of 256 × 256 pixels. Due to its high resolution and detailed spatial structures, UCM is widely used for evaluating classification algorithms in remote sensing.

2) Aerial Image Dataset: The AID dataset is a large-scale benchmark designed to improve model generalization under more complex scene variations. It comprises aerial images collected from various countries, containing between 220 and 420 RGB images per class at 600 × 600 pixel resolution, totaling 10 000 images. The dataset emphasizes diversity across geography, seasonality, and land use types.

3) NWPU-RESISC45 (NWPU): NWPU-RESISC45 is among the largest datasets for remote sensing scene classification, featuring 45 scene classes with 700 high-resolution images (256 × 256 pixels) per class, amounting to 31 500 images. Its rich intraclass variations in scale, orientation, and context make it a challenging benchmark for evaluating the scalability and robustness of classification models.

4) Universal Adversarial Examples in Remote Sensing: The UAE-RS dataset is specifically curated for studying adversarial robustness in remote sensing classification. It includes adversarial examples generated across multiple datasets, such as UCM and AID, using a variety of attack strategies. By simulating realistic security threats, UAE-RS provides a critical benchmark for assessing the resilience of defense algorithms against adversarial perturbations.

Collectively, these four datasets encompass diverse scales, data distributions, and attack scenarios, offering a comprehensive evaluation environment for LDP-Diff. Extensive experiments across them enable a thorough assessment of the model's adversarial purification capability, robustness, and generalization performance.

B. Experimental Setting

1) Classifiers: In this study, we employ a comprehensive set of mainstream deep learning classification models, including AlexNet [24], VGG16 [55], ResNet (18, 50, 101) [58], DenseNet (121, 201) [59], Inception-v3 [60], and RegNetX-400MF [61]. These models are pretrained and evaluated on three representative remote sensing scene classification datasets: UCM, AID, and NWPU. The selected architectures span from early convolutional neural networks to deeper residual networks and densely connected networks, incorporating advanced architectural innovations such as multiscale

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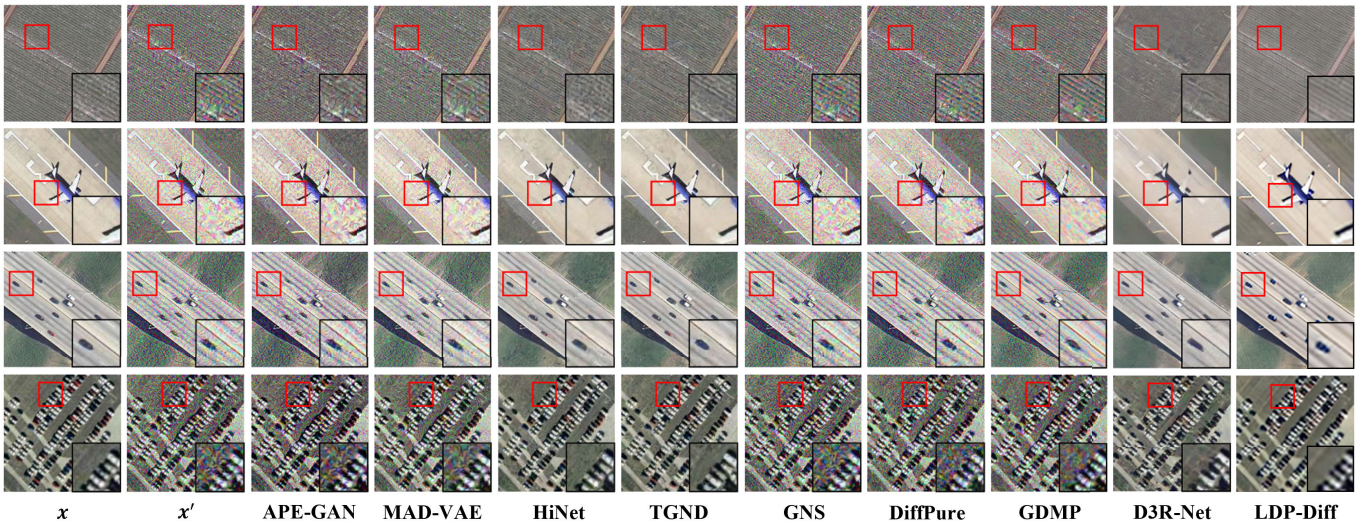


Fig. 3. Adversarial purification effects of various defense methods under UAE-RS attack on UCM dataset.

TABLE I
CLASSIFICATION ACCURACY (%) OF ALL MODELS ON THE UCM, AID, AND NWPU DATASETS

Model	UCM	AID	NWPU
AlexNet	90.10	89.62	89.93
Vgg16	91.90	92.28	92.33
ResNet18	96.10	95.08	94.29
ResNet50	97.24	91.26	93.73
ResNet101	96.48	92.50	93.38
DenseNet121	97.05	95.36	94.60
DenseNet201	97.71	95.41	95.16
Inception-v3	95.81	93.52	93.71
RegNetX-400MF	95.90	94.52	94.93

feature extraction and structural efficiency optimization. The classification accuracies of all models on the test sets of the aforementioned datasets are summarized in Table I.

To rigorously evaluate the robustness of our proposed LDP-Diff model under unseen adversarial attacks, we train the model using adversarial examples generated by UAE-RS and test its performance against a suite of widely adopted attack methods. For a fair comparison, we adopt the same experimental settings as those used in UAE-RS, UAD-RS, and D3R-Net. The considered attacks include FGSM, IFGSM, CW, Jitter, Mixcut, Mixup, and TPGD. FGSM introduces a one-step perturbation along the sign of the input gradient; IFGSM extends this by performing iterative updates to intensify the perturbation; CW attack formulates an optimization problem to generate highly deceptive adversarial examples; Jitter applies small random spatial perturbations to compromise model stability; Mixcut and Mixup construct hybrid examples through image patch blending and example interpolation, thereby enhancing attack transferability; and TPGD adopts a multistep optimization strategy driven by KL divergence to craft stronger and more effective adversarial perturbations.

2) Implementation Details: All experiments were conducted on an NVIDIA A6000 GPU server equipped with 50 GB of VRAM, implemented using the PyTorch framework. During training, LDP-Diff was trained for 200 epochs with the Adam optimizer, which was adopted with an initial learning rate of 1×10^{-4}, and a weight decay of 1×10^{-8} was applied to mitigate overfitting. Input images were initially resized to $3 \times 256 \times 256$ and then center-cropped to $3 \times 224 \times 224$ before being fed into the CLIP model. Dataset-specific preprocessing strategies were employed to accommodate the characteristics of different remote sensing datasets. For the proposed LDP-Diff model, the length of the learnable text prompt vector was set to $N = 64$, with hyperparameters $\gamma = 0.4$ and $\eta = 2$. Furthermore, to balance the contributions of different loss terms during training, the weighting coefficients in the classification-level loss were set as $\xi_1 = 2$, $\xi_2 = 0.1$, and $\xi_3 = 0.1$.

C. Defense Performance on UAE-RS Attack

In this section, we conduct a systematic evaluation of the proposed LDP-Diff method in terms of its defense performance against adversarial examples generated by UAE-RS on the UCM and AID remote sensing datasets. To ensure a comprehensive comparison, we benchmark LDP-Diff against a range of representative and SOTA adversarial purification methods. For the sake of reproducibility and fairness, we adopt the data partitioning protocol used in D3R-Net, whereby each dataset is randomly divided into 50% for training and 50% for testing.

1) Qualitative Experiment: The qualitative comparisons in Figs. 3 and 4 evaluate the reconstruction performance of various defense strategies against UAE-RS adversarial examples on the UCM and AID datasets. Generative approaches such as APE-GAN and MAD-VAE partially suppress perturbations but often introduce structural artifacts and texture distortions, compromising scene consistency. Advanced methods like HiNet, TGDN, and GNS exhibit local texture retention yet suffer

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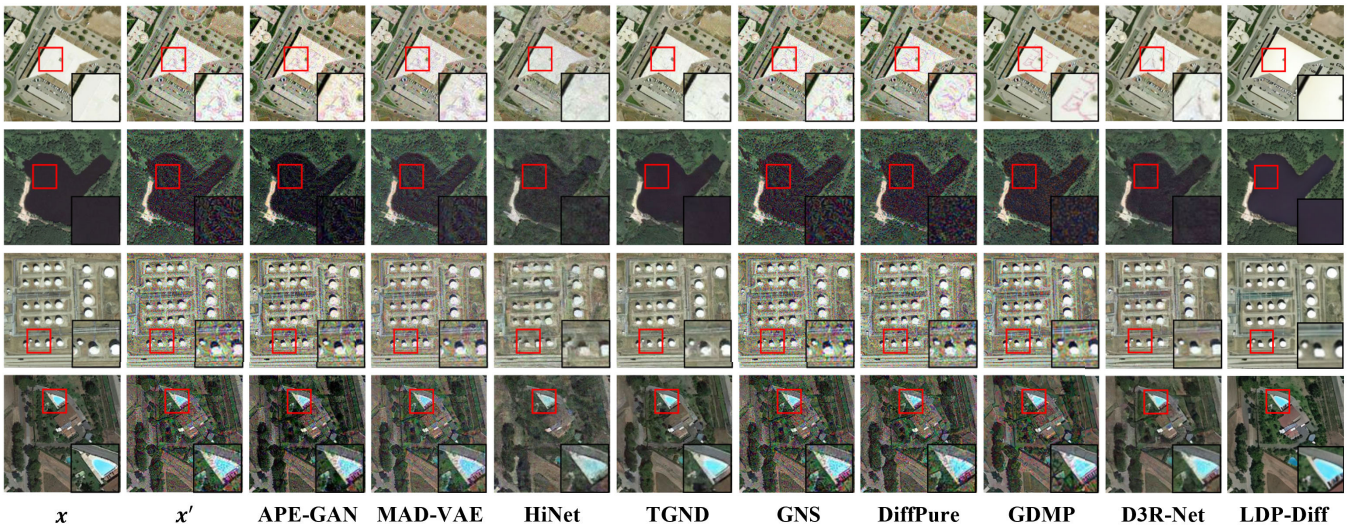


Fig. 4. Adversarial purification effects of various defense methods under UAE-RS attack on AID dataset.

TABLE II
QUANTITATIVE COMPARISON OF ADVERSARIAL DEFENSE METHODS FOR SCENE CLASSIFICATION ON UCM DATASET UNDER UAE-RS ATTACK

Model	None	APE-GAN	MAD-VAE	HiNet	TGDN	GNS	DiffPure	GDMP	D3R-Net	LDP-Diff
VGG16	19.52	39.33	37.14	57.24	68.67	32.10	37.24	62.67	77.48	**80.57**
Inception-v3	30.19	58.19	66.10	64.76	74.48	43.14	46.00	63.62	**83.06**	81.29
ResNet18	2.95	10.57	14.86	4.10	9.90	9.43	25.33	41.24	**87.43**	85.52
ResNet50	25.53	27.14	33.62	52.38	65.81	25.81	42.57	53.14	81.30	**84.71**
ResNet101	28.10	44.67	52.76	56.48	69.43	50.67	66.48	70.86	82.56	**85.24**
DenseNet121	17.14	32.00	44.00	45.81	61.52	25.62	35.14	65.33	83.37	**87.24**
DenseNet201	26.38	37.90	41.43	51.33	64.95	41.14	57.14	68.76	80.27	**85.90**
RegNetX-400MF	27.33	56.00	55.05	56.67	66.38	44.57	49.81	63.52	77.20	**81.38**
avg	22.14	38.23	43.12	48.60	60.14	34.06	44.40	61.14	81.58	**83.98**

from cross-domain instability and model inconsistency. These issues are especially pronounced in the more heterogeneous AID dataset, as evidenced by structural loss and boundary misclassification. Diffusion-based defenses (e.g., DiffPure and GDMP) offer better noise suppression but are limited by sampling variance, leading to output inconsistency across examples.

In contrast, the proposed LDP-Diff achieves consistently superior restoration quality. It preserves high-frequency edge details and semantic structures while effectively removing adversarial perturbations, as shown in the magnified regions. This robustness stems from integrating learnable prompts and the EGLF loss, enabling semantically aligned, efficient purification. Consequently, LDP-Diff delivers both stable visual recovery and enhanced classification discriminability.

2) Quantitative Experiment: As shown in Table II, under the no-defense (None) condition, all classifiers experience a substantial drop in classification accuracy when subjected to UAE-RS adversarial perturbations, clearly revealing the extreme vulnerability of remote sensing models to unseen attacks. Traditional purification-based approaches, such as APE-GAN and MAD-VAE, offer only marginal and unstable improvements across multiple classifier architectures. In contrast, the proposed LDP-Diff consistently achieves the highest classification accuracy across all eight classifiers, with an average accuracy of 83.98%, outperforming existing defense methods. For instance, on ResNet50, LDP-Diff achieves an accuracy of 84.71%, representing a 3.41% improvement over D3R-Net. The AID dataset, characterized by more complex category structures and greater domain shifts, poses a more significant challenge for defense methods regarding generalization. As indicated in Table III, in the absence of defense, the classification accuracies of most models drop below 10%, with ResNet18 falling to as low as 1.28%. Among all evaluated methods, LDP-Diff consistently achieves the best performance across all classifiers, with an average accuracy of 72.95%, yielding an improvement of approximately 8.96% over D3R-Net. Notably, on DenseNet201, LDP-Diff reaches 76.84%, significantly outperforming D3R-Net's 67.32%, indicating strong cross-model robustness. Furthermore, LDP-Diff demonstrates superior resilience against nongradient-based attacks such as MixCut and TPGD, underscoring its robustness in the face of more diverse and complex adversarial scenarios. This advantage stems from incorporating a learnable prompt mechanism and a semantically aware diffusion modeling strategy, enhancing image purification robustness and classification consistency.

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TABLE III
QUANTITATIVE COMPARISON OF ADVERSARIAL DEFENSE METHODS FOR SCENE CLASSIFICATION ON AID DATASET UNDER UAE-RS ATTACK

Model	None	APE-GAN	MAD-VAE	HiNet	TGDN	GNS	DiffPure	GDMP	D3R-Net	LDP-Diff
VGG16	11.88	19.60	26.00	47.78	51.34	17.92	18.50	41.92	56.84	**73.13**
Inception-v3	21.54	20.94	31.02	53.16	65.14	20.82	24.04	35.40	64.75	**65.86**
ResNet18	1.28	4.66	7.56	4.60	24.22	7.34	15.72	30.56	62.16	**75.49**
ResNet50	10.74	17.36	20.02	41.10	60.66	20.72	27.98	41.08	65.38	**72.69**
ResNet101	11.56	22.80	25.50	45.04	66.90	21.76	25.04	45.54	63.92	**75.08**
DenseNet121	10.30	19.84	22.56	39.28	58.16	16.66	26.42	43.36	65.38	**78.60**
DenseNet201	14.22	17.02	19.82	44.12	65.60	17.24	21.90	43.70	67.32	**79.84**
RegNetX-400MF	23.20	19.06	23.46	51.28	64.78	23.80	25.58	43.02	66.15	**77.92**
avg	13.09	17.66	21.99	40.80	57.10	18.28	23.15	40.57	63.99	**74.83**

Both qualitative and quantitative results demonstrate that LDP-Diff substantially outperforms existing adversarial purification methods, offering enhanced classifier robustness and strong generalization across diverse models and domains. This is attributed to the improved generalization capacity of the learnable defense prompts and EGLF loss, which jointly enforce consistency in pixel-level fidelity and semantic alignment during purification. Notably, experimental results on the UCM and AID datasets reveal that performance gains become more pronounced as the volume of training data increases, indicating that LDP-Diff can more effectively exploit richer data distributions to enhance adversarial purification and classification robustness.

D. Defense Performance on Various Invisible Attacks

This section systematically evaluates the generalization robustness of the proposed LDP-Diff framework against a wide range of imperceptible adversarial attacks. Extensive experiments are conducted on two benchmark remote sensing scene classification datasets, UCM and AID, employing multiple mainstream DNN classifiers. LDP-Diff is compared against a suite of classical and SOTA defense methods.

As reported in Table IV, on the UCM dataset, all models exhibit substantial degradation in classification accuracy under various attack scenarios when no defense is applied, with the average accuracy plummeting to 14.68%, thereby underscoring the pronounced vulnerability of remote sensing models to adversarial perturbations. Classical defense techniques such as ARN and NRP yield marginal improvements under gradient-based attacks (e.g., FGSM and CW). Still, their robustness deteriorates markedly when exposed to nongradient-based attacks such as Mixup and TPGD, indicating limited stability. PSGAN occasionally yields acceptable performance, yet suffers from inconsistency across attack types. In contrast, methods like TGDN and UAD-RS exhibit relatively better generalizability, maintaining higher accuracy under more substantial perturbations such as CW and TPGD. Remarkably, LDP-Diff achieves the highest overall average accuracy across all classifiers and attack variants, significantly outperforming existing defense baselines.

A consistent trend is observed on the AID dataset, as illustrated in Table V. Due to more pronounced domain shifts between training and testing distributions, AID imposes stricter requirements on the generalization ability of defense models. In the absence of defense, the average accuracy drops sharply to 11.43%, revealing the severity of invisible attacks in cross-domain remote sensing scenarios. While most baseline methods suffer significant performance declines, LDP-Diff consistently secures a leading position, achieving an average accuracy, surpassing competitive baselines such as UAD-RS and PSGAN. The advantage of LDP-Diff is particularly evident under nongradient-based attacks, including CW, TPGD, and Mixcut, demonstrating its strong generalization across both attacks and domains.

LDP-Diff demonstrates remarkable robustness and generalizability across diverse classifiers, attack methods, and datasets. Its success is primarily attributed to LDP and EGFL, which enhance the model's ability to perceive and suppress adversarial perturbations. Its superior performance under high-intensity perturbations highlights the effectiveness of the semantically guided diffusion-based purification strategy, which facilitates robust reconstruction of clean features while effectively suppressing adversarial noise.

E. Defense Performance on NWPU Dataset

To comprehensively evaluate the robustness of LDP-Diff in remote sensing image scene classification, we further assessed its defense performance on the NWPU dataset. As shown in the qualitative results in Fig. 5, LDP-Diff significantly improved image restoration quality when exposed to adversarial attacks. The method effectively mitigates adversarial perturbations while preserving the original texture and structure of the image, thereby reducing artifacts and noise introduced by the adversarial disturbances, demonstrating its powerful image recovery capabilities. Furthermore, the radar chart analysis further validates LDP-Diff's superior performance across various metrics. As depicted in Fig. 6, LDP-Diff's overall defense capability outperforms D3R-Net under multiple adversarial attacks, particularly maintaining high classification accuracy and robustness when dealing with complex model structures. The qualitative results and radar chart analysis demonstrate that LDP-Diff exhibits strong defense capabilities in complex environments on the NWPU dataset. In conclusion, LDP-Diff excels in remote sensing image classification tasks and demonstrates remarkable adaptability and robustness across various adversarial attack

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TABLE IV
QUANTITATIVE COMPARISON OF ADVERSARIAL DEFENSE METHODS FOR SCENE CLASSIFICATION ON UCM DATASET UNDER VARIOUS ATTACKS

Defense	Victim DNNs	Attack on UCM dataset							*avg*
		FGSM	IFGSM	CW	Jitter	Mixcut	Mixup	TPGD	
None	AlexNet	10.95	0.00	0.00	3.05	37.05	13.14	30.48	13.52
	DenseNet121	56.67	1.62	1.14	12.28	3.05	1.05	35.33	15.88
	ResNet18	33.90	0.00	0.00	6.28	8.28	3.62	35.81	12.56
	RegNetX-400MF	34.24	0.02	0.00	4.92	23.66	24.76	29.68	16.75
ARN	AlexNet	61.33	59.62	59.81	62.10	63.80	64.57	63.14	62.05
	DenseNet121	63.52	61.05	63.62	60.85	48.28	49.52	62.38	58.46
	ResNet18	58.76	57.33	58.95	57.33	47.24	50.67	58.95	55.60
	RegNetX-400MF	58.76	57.33	58.95	57.33	47.24	50.67	58.95	55.60
NRP	AlexNet	35.05	0.00	33.43	0.00	13.71	45.33	33.71	23.03
	DenseNet121	78.00	19.24	70.10	20.19	1.43	3.43	76.96	38.48
	ResNet18	67.52	5.24	64.38	6.19	4.57	9.62	38.76	28.04
	RegNetX-400MF	75.42	28.38	72.86	28.86	29.52	41.52	44.19	45.82
PSGAN	AlexNet	**84.19**	36.95	36.76	71.52	46.38	29.42	41.71	49.57
	DenseNet121	87.90	29.71	30.48	80.10	6.67	4.28	53.52	41.81
	ResNet18	84.10	38.76	40.19	80.19	33.14	25.90	67.03	52.76
	RegNetX-400MF	85.33	44.48	49.90	83.14	65.33	54.00	55.90	62.58
TGDN	AlexNet	46.28	30.48	31.05	53.14	52.10	44.28	48.38	43.67
	DenseNet121	77.81	64.76	62.57	79.43	29.62	30.10	74.48	59.82
	ResNet18	71.14	63.43	64.10	74.76	50.38	46.19	72.95	63.28
	RegNetX-400MF	66.19	60.10	60.19	66.67	68.10	61.14	64.95	63.91
UAD-RS	AlexNet	65.43	53.33	54.10	70.10	51.24	66.28	64.86	60.76
	DenseNet121	81.71	68.57	68.19	82.79	40.48	39.05	75.33	65.16
	ResNet18	78.00	66.48	66.86	80.48	54.19	48.95	75.52	67.21
	RegNetX-400MF	84.28	75.81	75.81	84.19	72.57	62.19	80.28	76.45
LDP-Diff	AlexNet	81.33	**59.33**	**60.00**	**79.38**	**69.86**	**71.52**	**74.57**	**70.86**
	DenseNet121	**88.67**	**84.29**	**84.10**	**87.62**	**65.52**	**66.95**	**87.05**	**80.60**
	ResNet18	**87.24**	**85.62**	**84.43**	**86.67**	**69.24**	**68.43**	**87.09**	**81.25**
	RegNetX-400MF	**86.33**	**83.48**	**83.05**	**87.95**	**86.76**	**76.10**	**85.62**	**84.18**

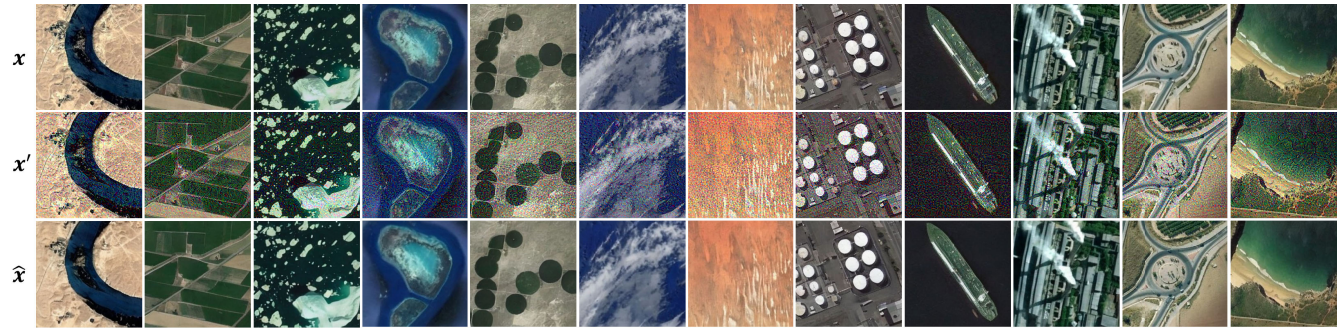


Fig. 5. Qualitative defense results of LDP-Diff on NWPU dataset.

scenarios, providing an effective solution for the security defense of remote sensing images.

F. Comparison of Semantic Attention Based on Grad-CAM

Fig. 7 shows the class activation map (Grad-CAM) [62] visualization results of typical remote sensing images on the three datasets of UCM, AID and NWPU in the natural state, after adversarial perturbation, and after LDP-Diff purification. The three columns of each image set correspond to the natural, antagonistic, and purified examples, and the green bars indicate their Top-3 prediction categories and confidence levels. The visualization results are based on the last layer of convolutional features of the baseline model ResNet-18, reflecting the migration trend of the model's regions of interest with respect to the input. As seen in the natural examples (first column), the model is able to accurately focus on semantically critical regions. However, in the adversarial example (second column), the perturbation leads to a significant shift in the

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TABLE V
QUANTITATIVE COMPARISON OF ADVERSARIAL DEFENSE METHODS FOR SCENE CLASSIFICATION ON AID DATASET UNDER VARIOUS ATTACKS

Defense	Victim DNNs	Attack on AID dataset							*avg*
		FGSM	IFGSM	CW	Jitter	Mixcut	Mixup	TPGD	
None	AlexNet	12.82	0.00	0.00	2.32	28.36	3.64	24.10	10.18
	DenseNet121	46.08	0.02	0.02	4.74	0.02	0.02	24.44	10.76
	ResNet18	15.30	0.00	0.00	3.68	2.64	0.14	34.50	8.04
	RegNetX-400MF	34.24	0.02	0.00	4.92	23.66	24.76	29.68	16.75
ARN	AlexNet	39.32	37.96	40.42	38.06	53.66	51.14	39.30	42.84
	DenseNet121	49.62	47.94	49.68	47.92	29.38	29.38	49.18	43.3
	ResNet18	38.36	37.30	38.58	37.86	41.48	42.18	38.88	39.23
	RegNetX-400MF	42.52	41.30	43.72	41.30	30.16	33.52	44.38	39.56
NRP	AlexNet	31.46	0.12	29.26	0.20	4.56	31.16	27.64	17.77
	DenseNet121	67.28	9.64	53.82	10.92	0.18	0.16	30.32	24.62
	ResNet18	40.18	3.10	40.52	3.40	0.32	2.88	35.62	18.00
	RegNetX-400MF	59.46	12.82	55.58	14.32	27.54	25.66	38.52	33.41
PSGAN	AlexNet	**79.96**	42.46	42.24	69.08	33.92	16.84	43.54	46.86
	DenseNet121	83.12	58.22	58.00	82.62	18.48	18.46	69.50	55.49
	ResNet18	80.52	61.26	60.86	81.76	33.16	29.60	72.60	59.97
	RegNetX-400MF	82.08	50.56	54.18	80.58	57.78	52.42	65.22	63.26
TGDN	AlexNet	26.42	18.24	18.36	28.78	22.88	18.28	27.30	22.89
	DenseNet121	41.96	34.08	33.86	41.94	14.46	14.18	38.74	31.32
	ResNet18	32.12	27.70	28.68	34.72	19.76	18.28	33.90	27.88
	RegNetX-400MF	34.44	30.14	30.66	35.84	35.24	32.16	33.78	33.18
UAD-RS	AlexNet	68.68	57.12	57.40	72.56	47.98	30.12	59.16	56.15
	DenseNet121	75.02	59.36	59.08	75.36	31.54	32.10	63.04	56.50
	ResNet18	66.52	56.44	57.34	69.28	34.84	34.50	62.16	54.44
	RegNetX-400MF	76.88	67.72	68.20	78.50	59.42	53.28	73.00	68.14
LDP-Diff	AlexNet	79.22	**62.82**	**62.68**	**75.80**	**62.76**	**57.10**	**68.86**	**67.03**
	DenseNet121	**85.16**	**72.22**	**71.98**	**82.76**	**64.50**	**64.08**	**80.00**	**74.39**
	ResNet18	**83.46**	**74.66**	**74.68**	**84.94**	**64.88**	**64.72**	**87.44**	**76.40**
	RegNetX-400MF	**83.88**	**79.30**	**79.54**	**85.50**	**74.72**	**69.24**	**82.20**	**79.20**

attention graph and the model focuses on irrelevant regions, exhibiting a clear inference failure. In contrast, the image purified by LDP-Diff (third column) effectively reconstructs the model's attention distribution, realigning the attention regions to the semantic core regions and significantly improving classification consistency and confidence. Meanwhile, the purified examples are highly consistent with the natural examples in terms of classification probability distribution, indicating that the purification not only resists the adversarial perturbation, but also maintains the integrity of the discriminant semantics.

In summary, the class activation graph visualization verifies the robustness and interpretability of LDP-Diff in multiple datasets and scenarios. It shows significant advantages in recovering the discriminant logic, reconstructing the semantic regions of interest, and improving the classification stability, which provides a strong support for the robust application of remote sensing images in safety-critical scenarios.

G. Evaluating Cross-Domain Performance

To systematically evaluate the cross-domain generalization capability of LDP-Diff in remote sensing image classification under adversarial conditions, we designed a series of cross-domain experiments based on three representative datasets: UCM, AID, and NWPU. Specifically, the model is independently trained on each dataset and then tested on the remaining two, simulating real-world scenarios where the training and testing distributions differ. This setup comprehensively assesses the model's transfer robustness across different domains. All adversarial examples are generated from UAE-RS, based on Mixcut-Attack crafted by the ResNet18 model, ensuring consistent attack intensity and experimental comparability.

As shown in Fig. 8, LDP-Diff consistently maintains good classification accuracy across all domains and classifier backbones. This confirms the robustness and generalizability of our proposed defense mechanism, which leverages learnable prompts and diffusion-based purification. Among the three source domains (UCM, AID, and NWPU), models trained on UCM exhibit noticeably lower average accuracies in cross-domain evaluations, particularly when tested on AID and NWPU. This phenomenon can be attributed to the limited diversity and scale of the UCM dataset, which contains fewer

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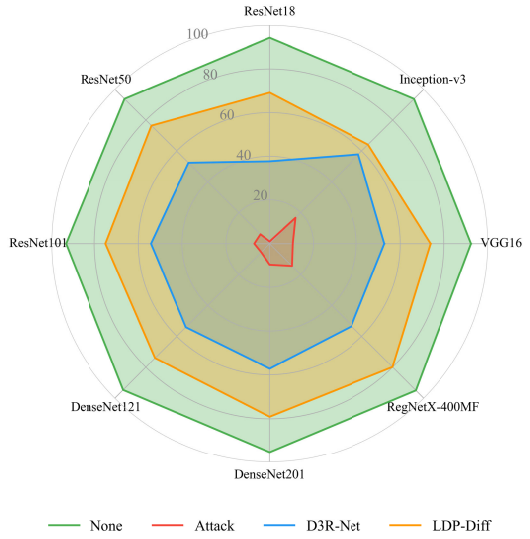


Fig. 6. Radar plot of classification accuracy on NWPU dataset across various classifiers. LDP-Diff demonstrates overall higher defense accuracy against UAE-RS attack compared to D3R-Net.

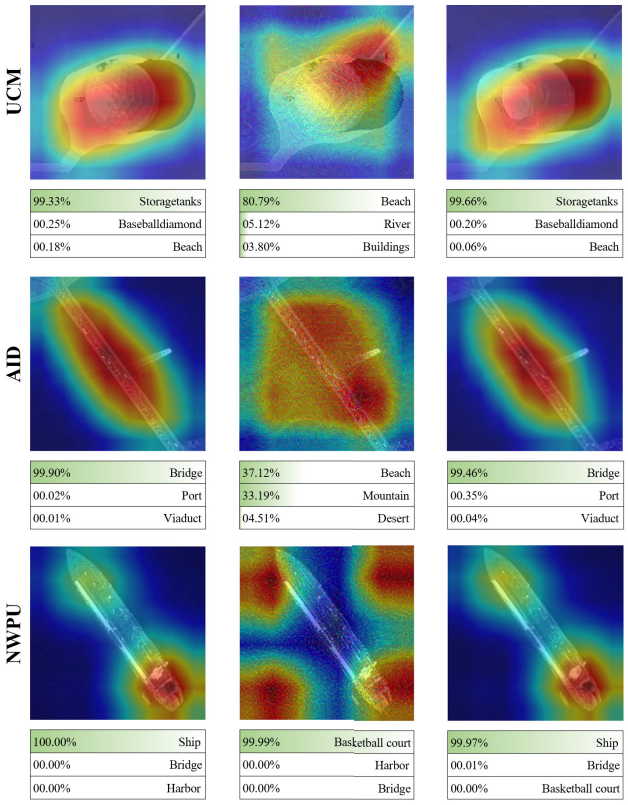


Fig. 7. Grad-CAMs of three RSIs on UCM, AID, and NWPU dataset. The first column represents natural RSIs, the second column is the adversarial RSIs produced by UAE-RS, and the third column represents the RSIs reconstruction by LDP-Diff. The green bars at the bottom of each RSI show the top three predicted classes and probabilities.

scene categories and examples than AID and NWPU. As a result, models trained on UCM are more prone to overfitting the source distribution and fail to capture transferable features that generalize well to broader target domains. In contrast, AID and NWPU, with their larger data volume and

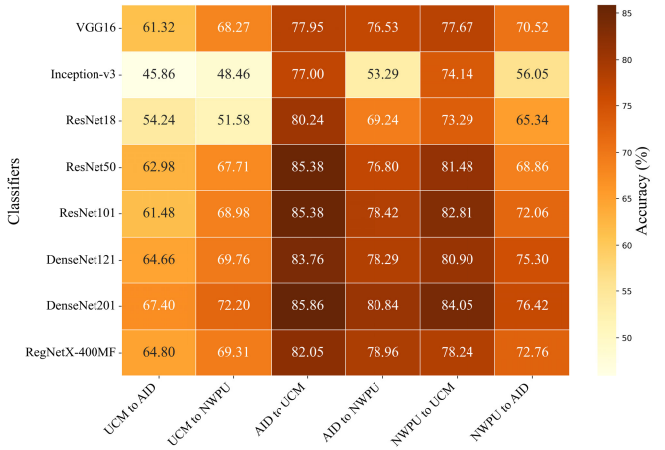


Fig. 8. Cross-domain generalization performance of LDP-Diff across UCM, AID, and NWPU datasets evaluated with diverse backbone classifiers.

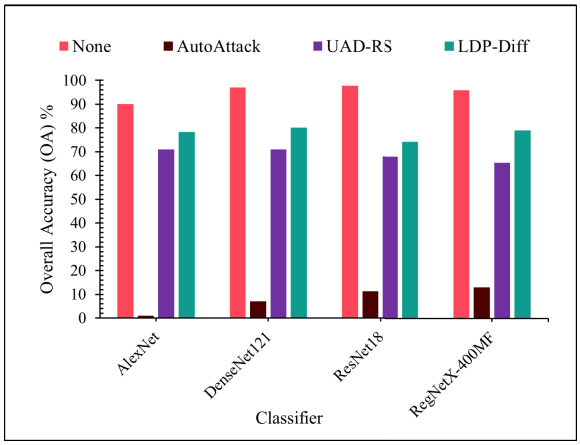


Fig. 9. Classification performance comparison of clean examples, adversarial examples generated by AutoAttack, and reconstruction examples by UAD-RS and LDP-Diff.

more diverse scene structures, promote the learning of more domain-invariant features, which benefits adversarial defense in unseen domains. Notably, the Inception-v3 consistently lags behind other classifiers in all cross-domain evaluations. This underperformance may stem from its architectural characteristics. Specifically, Inception-v3 relies heavily on multiscale convolutional filters and aggressive dimensionality reduction, which, although beneficial for clean-image classification, might amplify vulnerability to adversarial perturbations or overfit certain scale-specific features that do not generalize well across domains.

H. Defense Performance Against AutoAttack

To rigorously assess robustness against advanced adversarial threats, we evaluated LDP-Diff using AutoAttack, a parameter-free ensemble of gradient-based and gradient-free strategies that offers unbiased and stringent testing. Applied to all surrogate classifiers on the UCM dataset, as shown in Fig. 9, AutoAttack drastically degraded the performance of undefended models, exposing their vulnerability to strong, transferable perturbations. In contrast, LDP-Diff consistently

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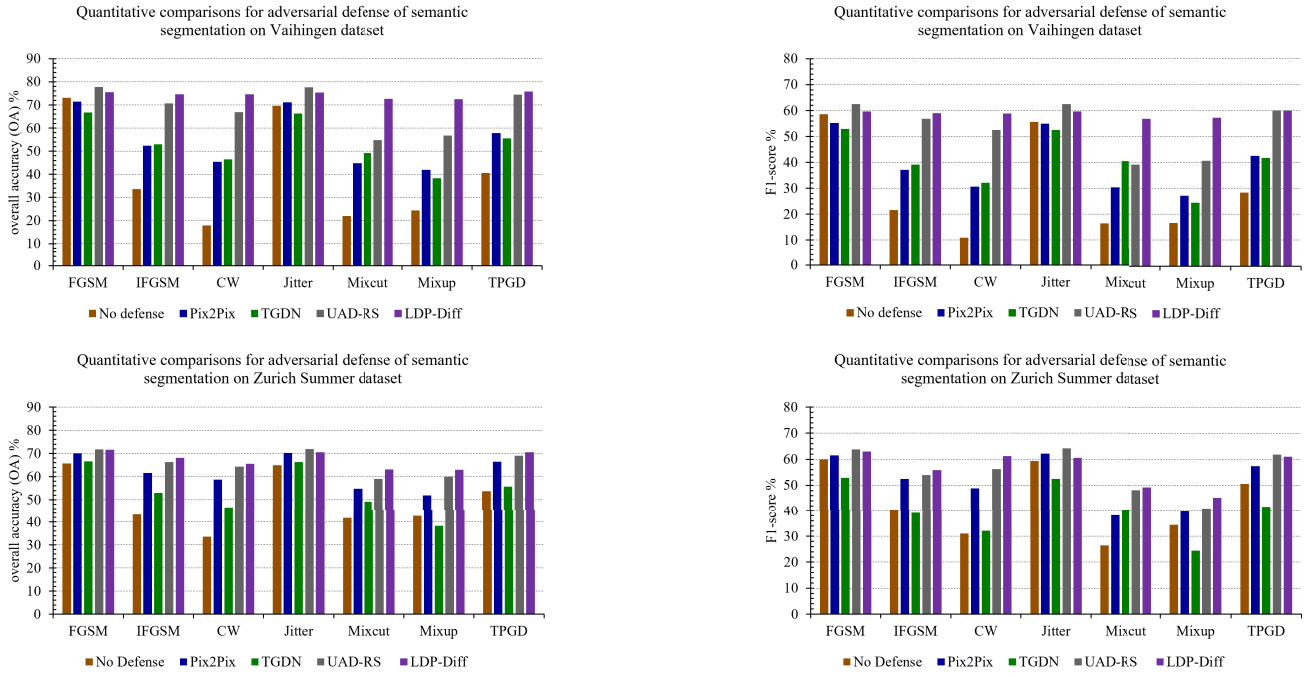


Fig. 10. Quantitative defense performance comparison of different methods on the vaihingen and zurich summer datasets under multiple adversarial attacks. Top row: overall accuracy (OA) and F1-score (F1) on the vaihingen dataset. Bottom row: OA and F1 on the zurich summer dataset.



Fig. 11. Visualizing LDP-Diff adversarial purification against APPA and CBA on RSOD dataset.

achieved higher robust accuracy and exhibited lower performance variance than UAD-RS and other baselines, confirming its reliable and transferable defense capability under the most rigorous evaluation settings.

I. Defense Performance on Semantic Segmentation

To validate the versatility of LDP-Diff beyond image classification, we conducted semantic segmentation experiments on the Vaihingen [63] and Zurich Summer [64] datasets using a U-Net [65] backbone fine-tuned on clean data. Adversarial examples were generated with the UAE-RS suite, and LDP-Diff was applied as a preprocessing purification step before segmentation. As shown in Fig. 10, compared with representative SOTA defenses under standard metrics (OA and F1-score), LDP-Diff consistently delivered superior robustness with lower variance across attacks and datasets. This stability, particularly evident on the complex Zurich Summer scenes, underscores LDP-Diff's ability to provide reliable, transferable, and semantically faithful defense for remote sensing segmentation tasks, reinforcing its suitability for real-world operational pipelines.

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TABLE VI
COMPARISON OF TIME COST FOR LDP-DIFF ON UAE-RS DATASET

Method	Time Cost (S/F)	Params (MB)	FLOPs
APE-GAN	0.03	17.26	40.20G
MAD-VAE	0.01	12.80	16.51G
HiNet	0.33	88.67	170.48G
TGDN	0.33	89.19	172.86G
GNS	8.21	552.66	1.11T
DiffPure	13.18	552.66	1.11T
GDMP	3.94	552.66	1.11T
D3R-Net	1.02	93.52	193.77G
LDP-Diff	0.21	17.05	597.38G

TABLE VII
EVALUATION RESULTS OF LDP-DIFF AND VARIOUS ADVERSARIAL PURIFICATION METHODS IN RSI ENHANCEMENT TASKS

Defense	L_∞	MSE	SSIM	PSNR	OA
None	0.0046	0.0009	0.8245	30.59	0.02
ARN	0.0164	0.0068	0.5338	22.71	57.33
NRP	0.0075	0.0026	0.6359	25.88	28.38
Pix2Pix	0.0107	0.0023	0.7777	26.73	6.50
PSGAN	0.0133	0.0666	0.1574	11.86	44.48
TGDN	0.0195	0.0840	0.3105	11.08	60.10
Diffusion	0.0199	0.0060	0.6690	24.06	75.81
LDP-Diff	0.0128	0.0038	0.6991	24.33	79.48

J. Defense Performance Against Background Attack

To assess LDP-Diff's efficacy against CBAs in remote sensing, we conducted supplementary experiments, rigorously reproducing APPA [35] and CBA [36] using their official codes for reproducibility. Following the APPA and CBA settings, we employed a pretrained YOLOv3 [66] model as the surrogate detector on RSOD dataset. Postpurification via LDP-Diff, patches were seamlessly merged via inverse stitching, preserving image quality for YOLOv3 detection. Fig. 11 illustrates natural examples, adversarial examples from CBA and APPA (causing detection failure), and LDP-Diff-purified samples, which restore semantic structures and enable accurate detection. These findings underscore LDP-Diff's robustness against complex, physically realizable perturbations and patch-based attacks, affirming its scalability for real-world remote sensing object detection.

K. Time Cost and Computational Complexity Analysis

In this section, we systematically evaluate the computational efficiency and resource consumption of the proposed LDP-Diff in comparison with SOTA adversarial purification methods. All evaluations are conducted on the UCM subset of the UAE-RS dataset to ensure consistency and fairness. Specifically, we report the average processing time per image (S/F), model size (Params), and computational complexity measured by floating point operations (FLOPs), as summarized in Table VI.

Regarding runtime efficiency, LDP-Diff achieves a significantly lower inference latency of 0.21 s/image, markedly outperforming diffusion-based baselines such as DiffPure, GNS, and GDMP. Regarding model complexity, LDP-Diff contains only 17.05 MB of parameters, which is considerably smaller than the large-scale diffusion frameworks and comparable to lightweight baselines such as APE-GAN and MAD-VAE. This compact model size makes LDP-Diff particularly well-suited for deployment in resource-constrained environments, including embedded systems and edge devices. Although LDP-Diff requires 597.38 G FLOPs, which is higher than that of D3R-Net and HiNet, it remains significantly more efficient than GNS and DiffPure. This demonstrates LDP-Diff's ability to achieve a favorable trade-off between purification accuracy, runtime, and computational cost. The model's modular design further enables semantic consistency during reconstruction while eliminating redundant computations typical in multistep denoising pipelines.

L. Image Enhancement-Based Evaluation Metrics

We employed various image enhancement evaluation metrics to comprehensively quantify the effectiveness of different adversarial defense methods on remote sensing images. Specifically, we selected L_∞-norm, mse, structural similarity index measure (SSIM), peak signal-to-noise ratio (PSNR), and overall accuracy (OA) as metrics to evaluate both the image restoration quality and its impact on classification performance. The L_∞-norm measures the maximum perturbation of an image, mse reflects the precision of image restoration, SSIM assesses the structural similarity of the image, PSNR is a commonly used metric for image quality, and OA directly evaluates the accuracy of the classifier on the purified image.

As shown in Table VII, methods like ARN and NRP perform poorly in terms of L_∞-norm and mse, indicating limited effectiveness in removing adversarial perturbations. In contrast, diffusion-based purification methods (e.g., Diffusion) outperform traditional methods in mse and SSIM. Our proposed LDP-Diff excels in all metrics. Specifically, LDP-Diff achieves impressive results with an mse of 0.0038 and SSIM of 0.6991 while showing significant improvements in PSNR and OA, demonstrating its strong capability in image quality restoration and classification accuracy enhancement.

Overall, LDP-Diff performs exceptionally well in the adversarial purification tasks for remote sensing images, especially in complex environments. It shows remarkable image restoration and classification abilities, which underscores its effectiveness and robustness in remote sensing image security.

M. Ablation Study

We conduct a detailed ablation study to evaluate the contribution of each component in LDP-Diff. As shown in Table VIII, the baseline without perceptual reconstruction loss $\mathcal{L}_{\text{mse}}$ performs poorly, indicating that pixel-level restoration alone cannot preserve semantic consistency. Adding perceptual loss $\mathcal{L}_{\text{PIPS}}$ improves reconstruction fidelity and segmentation accuracy by retaining fine textures and structural details. Incorporating the LDP further enhances robustness, confirming its role as an adaptive semantic prior that stabilizes diffusion outputs. $\mathcal{L}_{\text{EGLF}}$ provides additional gains by calibrating logits

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TABLE VIII
ABLATION STUDY EVALUATION RESULTS IN LDP-DIFF

$\mathcal{L}_{PIPS}$	LDP	EGLF	$\mathcal{L}_{CE}$	LoRA	UCM	AID	NWPU
					69.33	52.61	53.49
✓					76.24	68.03	67.65
✓	✓				80.38	71.52	70.86
✓	✓	✓			82.60	73.20	72.58
✓			✓		81.15	72.84	72.11
✓	✓	✓	✓	✓	83.98	74.83	73.94
✓	✓	✓	✓		82.29	75.08	73.22

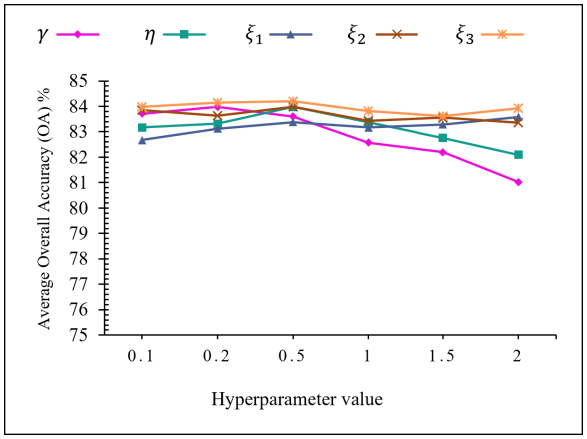


Fig. 12. Sensitivity analysis of hyperparameters.

based on uncertainty, reducing residual class bias. $\mathcal{L}_{\text{CE}}$ refines category boundaries and improves decision consistency, while LoRA-based fine-tuning offers further performance improvements with minimal computational cost. Finally, one-step diffusion achieves similar fidelity to multistep diffusion but with far greater efficiency, making it better suited for real-time applications. Together, these results demonstrate that each module contributes meaningfully, and their synergy underpins LDP-Diff's superior and stable performance across datasets.

N. Sensitivity Analysis of Hyperparameters

We conducted a sensitivity study of five key hyperparameters, γ, η, ξ_1, ξ_2, and ξ_3, that govern EGLF and multiobjective loss to assess LDP-Diff's robustness under varying configurations. As shown in Fig. 12, γ and η jointly control the strength and stability of semantic fusion, where extreme values either underutilize auxiliary information or overamplify uncertain logits, while moderate values provide balanced correction. ξ_1 regulates perceptual fidelity relative to pixel-level reconstruction, where insufficient weighting blurs fine details and excessive weighting may slightly compromise semantic alignment. ξ_2 and ξ_3, which scale EGLF loss and cross-entropy supervision, show minimal sensitivity, with moderate settings ensuring stable classification without overfitting. Across tested ranges, LDP-Diff maintains consistent performance with only minor fluctuations, confirming its robustness and practical deployability.

O. Limitations and Future Works

Although LDP-Diff achieves excellent overall classification accuracy, as shown in Fig. 13, its accuracy remains

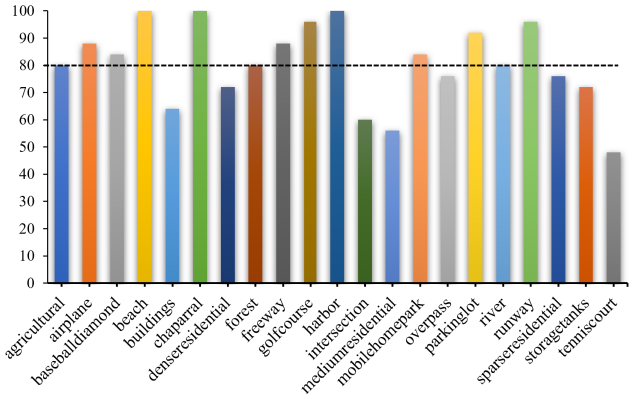


Fig. 13. Per-class classification accuracy of reconstruction examples by LDP-Diff on UCM dataset under UAE-RS attack using the VGG16 classifier.

relatively low for certain specific classes, such as "*tenniscourt*," "*mediumresidential*," and "*intersection*." These classes often exhibit complex features and high background similarity, challenging the model to distinguish them. For instance, "*tenniscourt*" shares structural similarities with other open-space classes like "*park*" or "*runway*." Meanwhile, "*mediumresidential*" and "*intersection*" are harder to differentiate due to the resemblance in building textures and street layouts, adversely impacting the model's recognition performance.

The ability of LDP-Diff to accurately recognize these specific complex classes still requires improvement. Future work will address interclass similarity by incorporating more refined class separation strategies, such as contrastive learning. This approach aims to optimize the decision boundaries between similar classes, enhancing recognition accuracy for these challenging cases.

V. CONCLUSION

RSI classification faces significant challenges from adversarial attacks, which can severely degrade model performance and compromise the reliability of downstream applications. To address these challenges, we propose LDP-Diff, a novel adversarial purification framework designed to enhance robustness in RSI scene classification. LDP-Diff integrates LDP with a one-step diffusion model, substantially improving example reconstruction quality while effectively preserving structural and textural details. The framework accelerates the purification process by employing a one-step diffusion model, eliminating the computational overhead associated with traditional multistep diffusion methods and thus enhancing processing efficiency. The proposed EGLF loss optimizes classification decisions within the semantic space, further boosting classification accuracy and adversarial robustness.

Extensive experiments on benchmark RSI datasets, including UCM, AID, NWPU, and UAE-RS, show that LDP-Diff consistently achieves superior defense performance compared with state-of-the-art adversarial purification methods

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under diverse adversarial attacks. Moreover, LDP-Diff exhibits robust cross-domain generalization, preserving high accuracy across datasets and validating its broad applicability. Compared to conventional diffusion-based approaches, LDP-Diff significantly accelerates the purification process without sacrificing reconstruction fidelity, improving overall time efficiency.

In summary, LDP-Diff delivers exceptional performance in RSI classification by advancing image recovery quality and reinforcing adversarial defense capabilities. Future work focused on refined class differentiation, multimodal learning, and model optimization holds promise to enhance its effectiveness further, paving the way for more resilient and precise adversarial defenses in RSI analysis.

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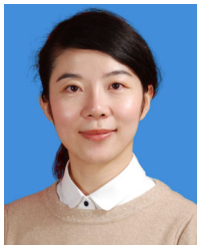
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